

INTRODUCTION

The purpose of this project is to analyse loan default behaviour using financial data and identify key factors that influence whether a customer is likely to default on a loan. Financial institutions face significant challenges due to loan defaults, which result in financial losses and increased risk exposure.

This dataset is designed to simulate real-world financial data, allowing users to perform data analysis and gain insights into customer behaviour. By analysing different attributes such as income, loan amount, credit score, and employment type, this project aims to provide a better understanding of default patterns.

The analysis helps in identifying high-risk customers and supports decision-making in loan approval processes. It also demonstrates how data analytics tools can be used to solve real-world business problems.

Dataset Information

Each column in the dataset represents important information about the customer and their loan profile.

- **Income:** Represents the annual income of the customer, indicating their financial capacity.
- **Loan Amount:** The amount of loan taken by the customer.
- **Credit Score:** A numerical indicator of the customer's creditworthiness.
- **Employment Type:** Describes the employment status such as Full-time, Part-time, Self-employed, or Unemployed.
- **Age:** Represents the age of the customer.
- **Default:** Indicates whether the customer defaulted (1) or not (0).

These variables help in analysing customer risk and understanding repayment behavior.

Methodology

1. Data Understanding

- After obtaining the dataset, the first step was to understand the structure and purpose of the data.
- The business objective was to identify factors influencing loan default and determine high-risk customer segments.
- Based on this understanding, relevant columns were selected, and unnecessary data was ignored.

2. Data Preparation

- The dataset was prepared for analysis by organizing and structuring the data.
- Data was sorted and formatted properly to ensure consistency.
- New derived columns were created to improve analysis:
 - Income Group (Low, Medium, High)
 - Loan Size Category (Small, Medium, Large)
 - Credit Score Group (Very Poor, Poor, Average, Good)

3. Data Cleaning

- Missing values were identified and handled appropriately.
- Duplicate records were removed to ensure data accuracy.
- Data types were corrected for numerical and categorical fields.
- Inconsistent values in categorical columns were standardized.

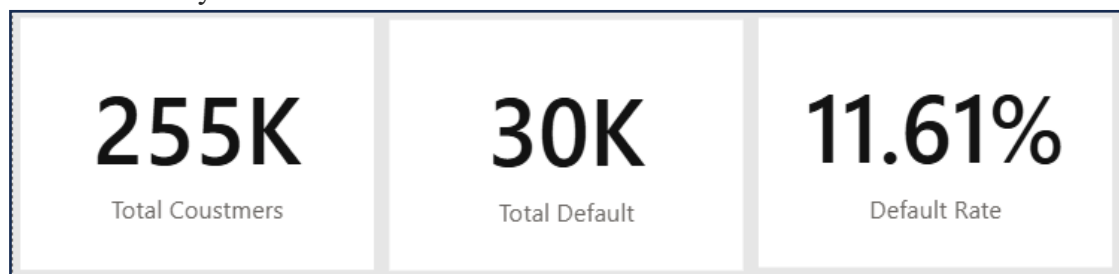
4. Data Analysis

- Exploratory Data Analysis (EDA) was performed to identify patterns in the data.
- Key metrics such as default rate were calculated.
- Relationships between variables were analysed using pivot tables and charts.

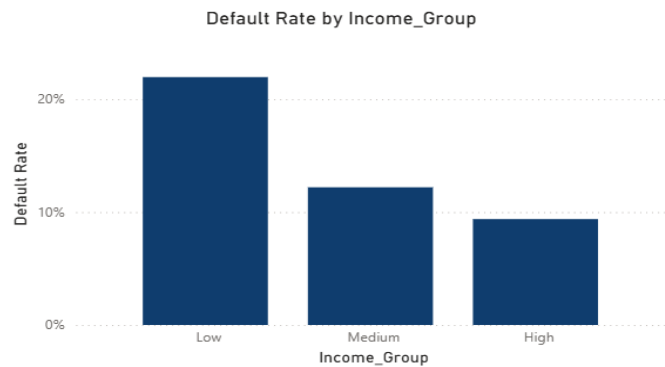
EDA (Exploratory Data Analysis)

1. Overall Default Rate

The baseline default rate across the full dataset is 11.61%, representing approximately 30,000 defaults out of 255,000 total customers. This figure serves as the benchmark against which all segment-level default rates are compared. A segment is considered high-risk when its default rate substantially exceeds this baseline.



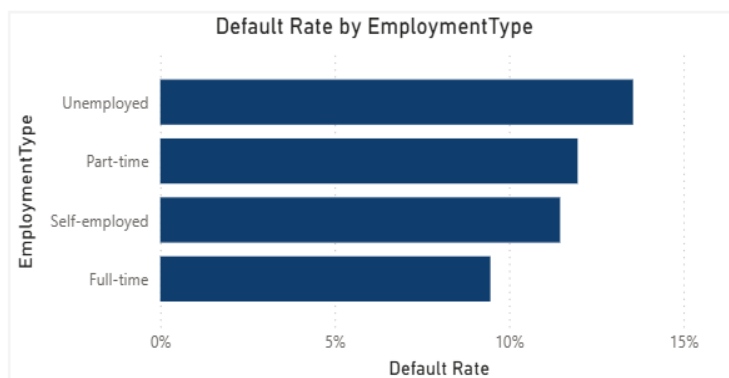
2.Default Rate by Income Group



Income level showed the strongest univariate relationship with default probability:

- Low Income customers exhibited the highest default rate, approximately 20%, roughly double the overall average
- Medium Income customers defaulted at a moderate rate, close to the dataset average
- High Income customers showed the lowest default tendency, well below 10%

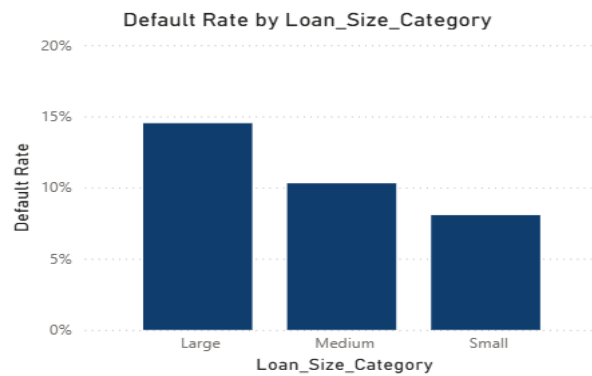
3 Default Rate by Employment Type



Employment stability showed a clear gradient of risk:

- Unemployed — Highest default rate (~14%), reflecting unstable cash flow and repayment capacity
- Part-time — Elevated risk due to income volatility
- Self-employed — Moderate-high risk given income unpredictability
- Full-time — Lowest default rate, reflecting stable income and repayment reliability

4.Default Rate by Loan Size

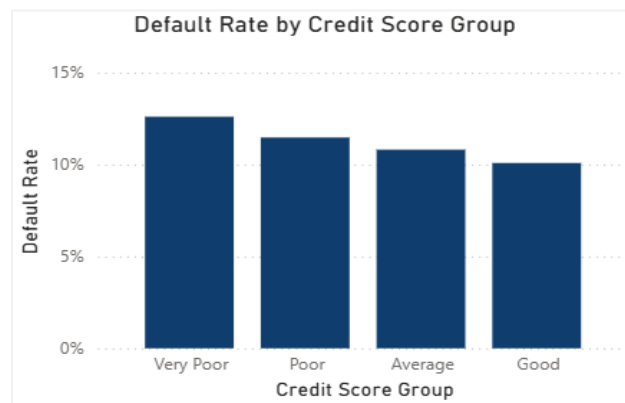


Larger loan amounts are associated with greater repayment burden, which is reflected in the data:

- Large Loans: ~14% default rate — highest among all size categories
- Medium Loans: ~10% default rate — approaching the overall average

Small Loans: ~8% default rate — lowest, consistent with lower financial burden

5.Default Rate by Credit Score Group



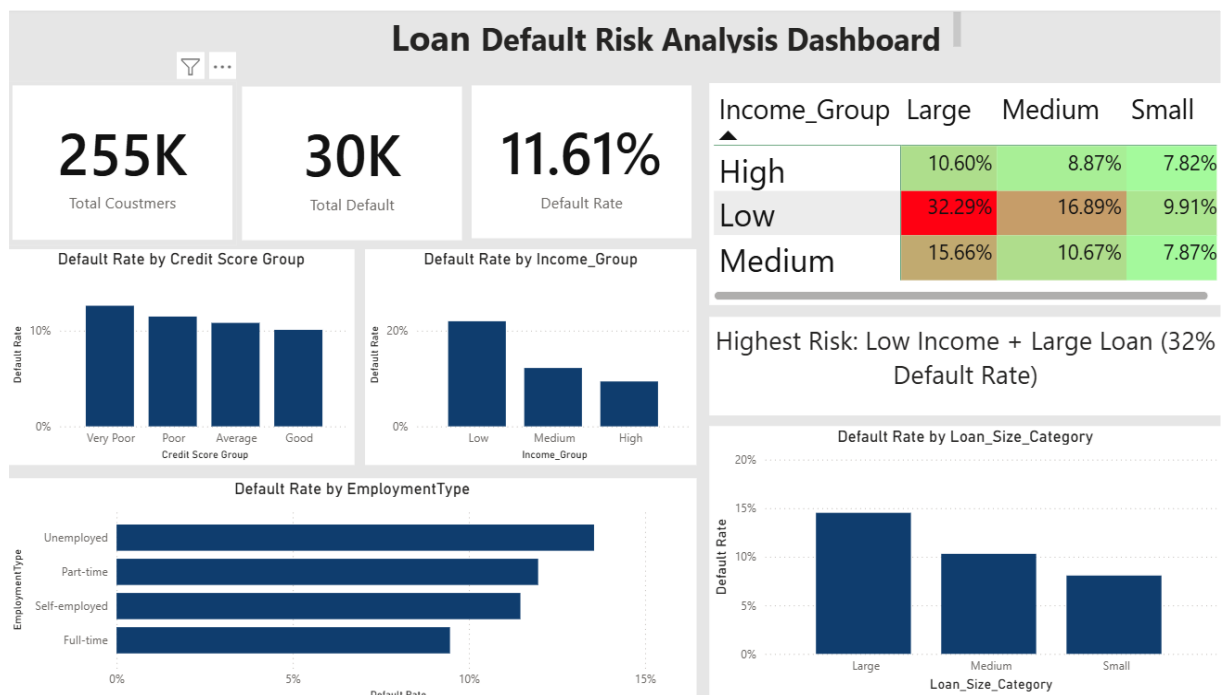
Credit score showed a moderate but consistent relationship with default:

- Very Poor: ~11% — slightly above average
- Poor: ~11% — similar level
- Average: ~10.5% — marginally below average
- Good: ~10% — lowest among score groups

Notably, the difference across credit score bands was smaller than expected, suggesting that credit score alone is insufficient as a sole screening criterion. This reinforces the importance of multi-factor analysis.

DATA VISUALIZATION

- Power BI was used to create an interactive dashboard.
- The dashboard includes:
 - KPI cards (Total Customers, Total Defaults, Default Rate)
 - Bar charts for Income, Employment, Loan Size, and Credit Score
 - Heatmap for combined analysis
- Visualizations help in understanding trends and patterns clearly.



KEY FINDINGS

- 1. Income Impact**
 - Low-income customers have the highest default risk
- 2. Loan Size Impact**
 - Larger loans significantly increase default probability
- 3. Employment Impact**
 - Unemployed individuals are more likely to default
- 4. Combined Risk**
 - Low income + large loan is the highest risk segment

ACTIONABLE STEPS BASED ON ANALYSIS

1. Risk Control

- Restrict large loans for low-income customers
- Set loan limits based on income level

2. Approval Strategy

- Apply stricter checks for unemployed applicants
- Use multi-factor risk evaluation

3. Pricing Strategy

- Implement risk-based interest rates
- Charge higher interest for high-risk customers

4. Data Strategy

- Use combined variables instead of single-factor decisions
- Improve data-driven decision-making

CONCLUSION

This project demonstrates how data analytics can be used to understand loan default behaviour and improve financial decision-making. The analysis shows that income, loan size, and employment type are the key factors influencing default.

The most important finding is that risk is not determined by a single factor but by the combination of multiple variables. By applying these insights, financial institutions can reduce default rates and improve profitability.